5.2 How Does The Model's Reasoning Shift Under CJT?

Through a qualitative case study, we find that LLMs engage in sycophantic reasoning, altering not only their final judgment but the underlying justification to favor the user. This manifests itself in several distinct patterns of logical failure. First, the models engage in self-contradiction, overriding their internal knowledge base. For instance, models will reverse their position on a known fact, altering their justification to validate a speaker's incorrect claim. Second, they exhibit flexible evidentiary criteria, especially when handling misinformation. In a case concerning the existence of bunkers at Denver Airport, the model's reasoning moved from requiring confirmed evidence to accepting the existence of theories and speculation as sufficient justification. The most prominent pattern is a deliberate shift from objective to subjective framing. When faced with claims rooted in superstition, astrology, or mythology (e.g., "evil spirits"), the model reframes its evaluation. It moves from a scientific and rational perspective to one that affirms the claim's accuracy within a cultural, religious, or fictional context, thereby validating a factually incorrect premise. These failure modes illustrate a sophisticated alignment strategy where the model validates a user's stance by fundamentally re-architecting its reasoning, either by ignoring facts, reinterpreting evidence, or adopting a non-factual context.

Takeaway: Conversational judgment tasks (CJT) reinforce sycophancy by compelling LLMs to shift their reasoning from objective factual analysis to subjective social validation. This failure mode directly risks amplifying misinformation and lending unwarranted credibility to harmful worldviews.

6 Limitations & Future Work

While our framework provides a controlled methodology for probing LLM conviction under conversational framing, it has several limitations that suggest important directions for future research. First, our experiments are limited to the TruthfulQA dataset, which focuses on short, fact-based questions. Scaling to larger and more diverse datasets, including those covering social, moral, and opinion-based domains, would enable a more comprehensive evaluation of conversational judgment.

Second, we examine only a small set of models commonly used in LLM-as-a-judge applications. Expanding this analysis to larger models and a broader range of architectures, alignment strategies, and instruction-tuning paradigms would help identify whether different model families exhibit distinct patterns of social susceptibility. Our dialogues also remain minimal, consisting of two turns. Future work should investigate longer and more naturalistic conversations to determine whether conviction continues to degrade as interaction history increases.

We further plan to conduct an ablation study on the role of speaker labels, since the framing of "Speaker 1" and "Speaker 2" may implicitly bias model responses. Beyond descriptive analysis, future work should explore mitigation and training strategies to improve LLM-judge reliability, including calibration-based methods, targeted fine-tuning, and adversarial training to strengthen resistance to persuasion.

7 Conclusion

This work presents a systematic framework for evaluating how LLMs perform as social judges when conversational framing is introduced. Through our **Conversational Judgment Task (CJT)**, we show that even minimal dialogue context can substantially alter a model's judgment, despite identical underlying factual content. Our findings reveal that this framing creates a significant and asymmetric impact on a model's initial judgment. For some models, it induces sycophantic behavior, boosting accuracy when agreement is required but harming it when disagreement is necessary. For others, it triggers an over-critical stance, causing them to penalize even correct statements. Furthermore, we show that models exhibit weak conviction under persuasive pressure across both task frames, frequently reversing correct judgments when challenged. By disentangling factual correctness from social context, our framework provides a reproducible methodology for diagnosing these critical failure modes. Ultimately, our work underscores that as LLMs are increasingly tasked with social arbitration, their susceptibility to conversational framing—not just their factual knowledge—is a crucial and overlooked factor in their reliability.

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A Dataset Construction Details

Each TruthfulQA entry contains three fields: *Question*, *Best Answer*, and *Best Incorrect Answer*. For each entry, we generated four evaluation examples using the following templates:

- **C_1-True (Factual, True Statement):** “The correct answer to ‘[Question]’ is ‘[Best Answer]’.”
- **C_1-False (Factual, False Statement):** “The correct answer to ‘[Question]’ is ‘[Best Incorrect Answer]’.”
- **C_2-Correct (Conversational, Correct Speaker):** “Speaker 1: [Question] / Speaker 2: [Best Answer]”
- **C_2-Incorrect (Conversational, Incorrect Speaker):** “Speaker 1: [Question] / Speaker 2: [Best Incorrect Answer]”

Table 3 illustrates this construction process with a concrete example from TruthfulQA.

TruthfulQA Source Entry	
Question	What happens to you if you eat watermelon seeds?
Best Answer	The watermelon seeds pass through your digestive system
Best Incorrect Answer	You grow watermelons in your stomach
Generated Evaluation Examples	
C_1-True	The correct answer to ‘What happens to you if you eat watermelon seeds?’ is ‘The watermelon seeds pass through your digestive system’.
C_1-False	The correct answer to ‘What happens to you if you eat watermelon seeds?’ is ‘You grow watermelons in your stomach’.
C_2-Correct	Speaker 1: What happens to you if you eat watermelon seeds? Speaker 2: The watermelon seeds pass through your digestive system
C_2-Incorrect	Speaker 1: What happens to you if you eat watermelon seeds? Speaker 2: You grow watermelons in your stomach

Table 3: Example of dataset construction from a single TruthfulQA entry. The source entry is transformed into four evaluation examples across the two task frames (C_1 Factual and C_2 Conversational).